220MWDC MINBU SOLAR POWER PLANT PROJECT (PHASE 1 50MWDC)

GEP (Myanmar) Co., Ltd

Project Title	220MWdc Minbu Solar Power Plant Project (Phase 1 50MWdc)
Version	1.0
Date of Issue	*14-12-2022*
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1 PROJECT DETAILS

1.1 Summary Description of the Project

220MWDC Minbu Solar Power Project (hereafter simplified as "the project") is a newly built grid-connected photovoltaic power plant with installed capacity of 220MWp, which is located in Minbu District, Magway Region, Myanmar. The project is divided into 4 phases with capacity of 50WMp for phase 1-3 and 70MWp for phase 4. Currently, Phase 1 has been operated since 27 September 2019 while phase 2 is under construction with expected COD in 2023. Phases 3 and 4 are expected to complete 1 and 2 years respectively from phase 2 COD's date (2024-2025). The project is owned by GEP (Myanmar) Co. Ltd. The existing scenario prior to the implementation of the project is the electricity being generated by Ministry of Electricity and Energy, Electric Power Generation Enterprise (EPGE), and the baseline scenario of the project is the same as the existing scenario. The installed capacity of the Project is about 220 MWdc which the main components of photovoltaic power plant or "solar field" consists of a large group of semiconductors technology-based silicon solar cells arranged in what is known as solar PV panel or solar module. Solar panels convert impinging sunrays (photons) to electrons. The electrons' flow generates direct current (DC) electricity which gets collected and channeled into an electronic device "inverter'' to invert the DC current into Alternating Current (AC) and step-up voltage by high voltage transformer and delivery to Myanmar national grid system which control by EPGE. The purpose of the project is to generate electricity by using renewable solar energy, and the electricity generated by the project will be delivered to EPGE for a 30 years' period ending 26 February 2049 to replace equivalent electricity generated by fossil fuel fired power plants connected to the EPGE. During the first crediting period, the annual average GHG emission reductions are estimated to be 48,722 t CO2, while the total GHG emission reductions during the first 10-year crediting period are estimated to be over 487,220 t CO2.

Contribution to sustainable development

The project promotes local sustainable development through the following aspects:

- The project activity will displace the power generation of fossil fuel power plants, reducing CO2, Sox and Nox emissions significantly, thus mitigating the air pollution and its adverse impacts on human health.
- Improvement of the fossil fuel dominated fuel mix of the electricity generation in the power grid by providing clean and renewable energy source and help to energy supply security.
- Promote application and diffusion of the innovative/creative solar PV technology in Myanmar through the demonstrative practice of the project activity.

1.2 Sectoral Scope and Project Type

The project is under sectoral scope 1 – Energy (renewable), ACM0002 – Large scale Consolidated Methodology: Grid-connected electricity generation from renewable sources Version 21.0

The project is not a grouped project.

1.3 Project Eligibility

The purpose of the project is to generate electricity by using renewable solar energy, and to replace equivalent electricity generated by fossil fuel fired power plants connected to the EPGE. The project currently generates 80,934 MWh/year with an Emission factor of 0.602, the project will approximately generate carbon emission reduction of 487,220 tCO2 for the 10-year crediting period. The project IRR is 6% which is lower than the benchmark's IRR of 16% (commercial lending rate). So, the project is not financial attractiveness. The project is located in Least Developed Country consisting in grid-connected electricity generation using solar power plant.

1.4 Project Design

The project includes a single location or installation only.

Eligibility Criteria

Not applicable

1.5 Project Proponent

Organization name	GEP (Myanmar) Co., Ltd
Contact person	Ronakrit Aussavachamrach
Title	Project Manager
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Email	ronakrit.a@geppower.net

1.6 Other Entities Involved in the Project

Not applicable

1.7 Ownership

The project is developed and owned by GEP (Myanmar) Co., Ltd. The ownership of the project shall be demonstrated through the following documents:

1. Power Purchase Agreement with Ministry of Electric Power Myanmar Electric Power Enterprise for sale of electricity by GEP (Myanmar) Co., Ltd
2. Memorandum of Association and Articles of Association of GEP (Myanmar) Co., Ltd

1.8 Project Start Date

27th September 2019

1.9 Project Crediting Period

This project adopts renewable crediting periods of 30 years (from 27/09/2019 to 26/02/2049), and the first 10-year crediting period is from 27/09/2019 to 26/09/2029 (both days included).

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

☐ <20,000 tCO_2e/year

☒ 20,000 – 100,000 tCO_2e/year

☐ 100,001 – 1,000,000 tCO_2e/year

☐ >1,000,000 tCO_2e/year

Project Scale	
Project	
Large project	

Year	Estimated GHG emission reductions or removals (tCO_2e)
Year 1 ((27/09/2019-31/12/2019))	12,181
Year 2	48,722

Year 3	48,722
Year 4	48,722
Year 5	48,722
Year 6	48,722
Year 7	48,722
Year 8	48,722
Year 9	48,722
Year 10	48,722
Year 11 (01/01/2028-26/09/2028)	36,542
Total estimated ERs	487,220
Total number of crediting years	10
Average annual ERs	48,722

1.11 Description of the Project Activity

The project will generate electricity by using renewable solar photovoltaic (PV) power and deliver to the Electricity Power Generation Enterprise or EPGE under the Ministry of Electricity and Energy (MOEE) Government of Myanmar, replacing equivalent electricity generated by fossil fuel fired power plants connected to the EPGE and therefore reducing GHG emissions.

The installed capacity of the Project is about 220 MWdc which the main components of photovoltaic power plant or “solar field” consists of a large group of semiconductors technology-based silicon solar cells arranged in what is known as solar PV panel or solar module. Solar panels convert impinging sunrays (photons) to electrons. The electrons’ flow generates direct current (DC) electricity which gets collected and channelled into an electronic device “inverter” to invert the DC current into Alternating Current (AC) and step up voltage by high voltage transformer and delivery to Myanmar national grid system which control by EPGE. The ~~expected~~ autual grid-in electricity in the first operation year is about 80,934~~000~~ MWh~~ (1st phase) and the accumulative decay rate during the whole lifetime in 30 years is 15%. ~~.T~~herefore, t~~he expected annual grid-in electricity in 30 years is 71,342~~313,684~~MWh~~, at >3,500 operational hours and the annual grid-in electricity in the first 10 years is 347,913.8MWh, at >3,500 operational hours~~. For the first 10-year crediting period, annual grid-in electricity of ~~347,913.8~~80,934 MWh is used for GHG reduction calculation.

The main equipment included:

1. Photovoltaic (PV), JETION 310- 320W. Warranty period is 12 years with power warranty of 90% for the first 10 years and 80% output for 25 years.

2. SMA 1x41MW 3 phase inverters

3. Medium volte 33kV Switch gear with below main characteristics of the devices;
   - Rated voltage / Operating voltage: 36kV / 33KV,
   - Rated short time withstand current Ik / Rated duration of short circuit: 20 kA/3s
   - AC (Internal Arc Classification): IAC A FL 20 kA/1s
   - Rated peak withstand current Ip: 63kA.
   - Rated normal current of the bus bar: 630 A
4. Power Plant Controller (PPC)
   - PPC will be installed in order to control the inverters. The solution will also manage the inverters will be integrated in the existing communication architecture with SCADA system.
5. Main Power transformers 230kV/33 kV with OLTC (50 MVA, ONAF) for phase 1- 3 and transformers 230kV/33 kV (70MVA ONAF) for phase 4.
6. 230 kV switchyard is designed conceptual by Double bus single breaker with 9 Feeders including;
   - BAY 1 is transmission line to Mann Substation
   - BAY 2 is transmission line to Ann Substation
   - BAY 3 is 230kV/33kV (50MVA) Power transformer No.1 (phase1)
   - BAY 4 is 230kV/33kV (50MVA) Power transformer No.2 (phase2)
   - BAY 5 is 230kV/33kV (50MAV) Power transformer No.3 (phase3)
   - BAY 6 is 230kV/33kV (70MVA) Power transformer No.4 (phase4)
   - • BAY 7 is Reactor no.1 (1x 25MVA)
   - • BAY 8 is Reactor no.2 (1x 25MVA)

- • BAY 9 is Bus coupler
- • Total 20 Disconnecting switch (DS) and 9 GCBs

7. Substation Equipment and specifications.

1.	Nominal Voltage	**11kV**	**33kV**	**230kV**
2.	Maximum Voltage Rating	12	36kV	245kV
3.	Power Frequency	50Hz	50Hz	50Hz
4.	Power Frequency Withstand Voltage • Phase-to-earth (r.m.s.)	28 kV	70kV	460kV
5.	Lightning Impulse Withstand Voltage • Phase-to-earth (peak)	75kV	170kV	1050kV
6.	Rated Short Time Current, 1s(r.m.s.)	31.5kA	40 kA	40 kA
7.	Switching Arrangement	Single bus	Single bus	Single bus
8.	Minimum insulator creepage distance	20mm/kV		
9.	Auxiliary Supply • AC • DC (Control & Protection)	• 400/230 V –3 phase 4 wire, grounded neutral • 220V-230 V		
10.	Operating voltage of control and protection equipment	85-110%		

8. Auxiliary Equipment

- Customer station for each phase.
- 33kV switchgear main bus.
- Control, Communication and Monitoring System
- SCADA system in CCB and substation room
- CCTV all PV fence area, Outside 230kV switchyard, CCB, Green area and all inverter stations.
- Safety Equipment
- Fire water supply system
- Water spray system
- Fire extinguishers
- Fire water storage tank
- Fire extinguishers system.

- Fire detection and alarm system
- Access road network

1.12 Project Location

The Project is located in Minbu District, Magway Region, Myanmar which is on the adjacent of Minbu - Ann road, 26 km. distance west of Minbu township of Magway Region. The access road to the Project will be about 1 km long.

The coordinates of the project are: 20°02'57.9"N 94°40'58.1"E (20.049426, 94.682802).

The location of the Project is shown in the map of Figure 1 and 2.

Figure 1: The location of the project

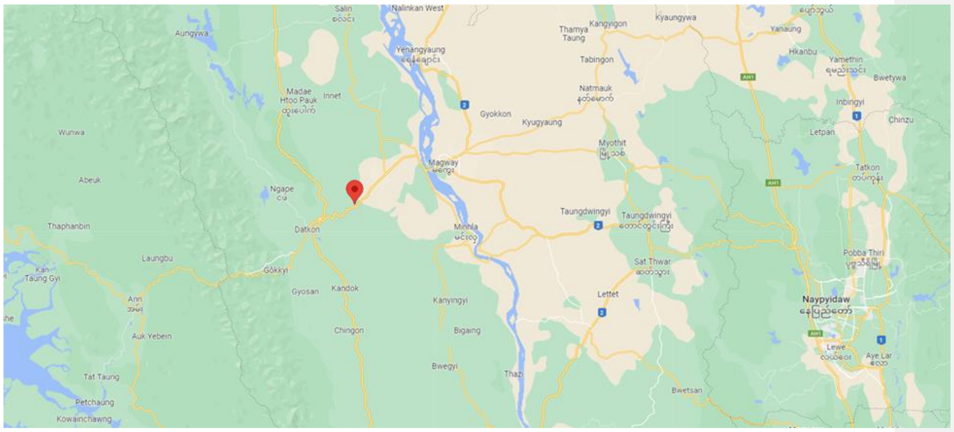

Figure 2: The location of the project

1.13 Conditions Prior to Project Initiation

The project is a greenfield power plant. The baseline scenario according to AM0002 Version 21 is "the electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants"

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

To date, there are no regulations policies preventing the implementation of the project activity. Senegal has prioritised the development of its use of the renewable energy sector, as the country has huge potential for solar and wind energy. The project is subject to the following legal and regulations:

1. Myanmar Investment Commission ("MIC") Permit
2. Permit to Trade (Directorate of Investment and Company Administration ("DICA") Permit);
3. Company Registration Certificate
4. Membership with the Union of Myanmar Federation of Chambers of Commerce and Industry (UMFCCI);
5. Export Import Certificate;
6. Construction permit;
7. Certificate from the Township Health Department under the Ministry of Health;

8. Labour/Safety related Permit
9. Environmental related Permit;
10. Permits required under the 2014 Electricity Law, Electricity Rules, and notifications and orders thereunder; and
11. State/Regional Permit

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project hasn't participated under CDM, GS or any other GHG Programs.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes ☒ No

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

- Social well-being: The project creates long term job opportunities during construction, operation, and maintenance states.
- Environment well-being: The project reduces CO2 emissions since it reduces the amount of fossil fuel used. In the case of "no project", the stated amount of electricity would be generated from fossil fuels and cause air pollution.

- Technology transfer: the project is the first solar power plant in Myanmar which can be viewed as a starter point for other project developers in Myanmar to develop solar power plant in Myanmar.
- GHG emission reduction: the project generates electricity from renewable energy which help avoiding using electricity generated from fossil-fueled power plant.

1.17.2 Sustainable Development Contributions Activity Monitoring

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7	7.b Expand infrastructure and upgrade technology for supplying modern and sustainable energy services for least developed country	Implemented activities to increase	No further changes this monitoring period	Increase sustainable energy by 40MWac
2)	8	8.5 Achieve full and productive employment and decent work for all	Implemented activities to increase	Generated job opportunities and income.	Current employment is 45 employees. Which includes people who lives in nearby village.
3)	9	9.1.1 Proportion of the rural population who live within 2 km of an all-season road	Implemented activities to increase	Donate the solar panel for 6KW to the museum for education purpose. The project gives 2% of its income before tax to support neighboring area each year.	The project plans to install solar roof top to Lay Pin Village High School for USD 4,398.92 for 2023 and the total budget for 2023 is USD 45,000

4)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By generating electricity from renewable sources. It helps to reduce CO2 emission by the same amount generating from fossil fuel.	Prevented the release of 152,884 thousand tonnes of carbon into the atmosphere during the monitoring period.

1.18 Additional Information Relevant to the Project

Leakage Management

According to AM0002 Version 21, no other leakage emissions are considered.

Commercially Sensitive Information

There is no commercially sensitive information for the project.

Further Information

Not applicable

2 SAFEGUARDS

2.1 No Net Harm

The project activity does not involve any major construction activity. It primarily requires the installation of solar PV panels, transformer stations and transmission lines. Solar PV project activity operations do not result in direct air pollution, noise pollution. Thus, there is no any significant impact due to implementation of project activity on air, water, soil quality and ambience are envisaged due to the project activity. There is no net harm from the project activity.

2.2 Local Stakeholder Consultation

Public consultations were held in Lay Pin and Zee Aign Village both villages. The public consultation in Lay Pin village was attended by 68 participants including member of parliament of Minbu Township Constituency-1, Regional Parliament, the general manager of GEP (Myanmar) company, the project developer, village-tract authorities, local elders, farmers and community members. It was held on September 12, 2016 morning and lasted for about two hours and a half. The consultation in Zee Aing village was attended by 38 participants who included farmers and community members from neighbouring villages. The GEP company officials were also present. The meeting was held in the afternoon of September 12, 2016. It also lasted about one hour and a half. In addition to two public consultations, key informant interviews were held with key stakeholders. They included five government personnel, 9 local authorities, community members, ~~farmers~~farmers, and religious leaders from Lay Pin Village and seven farmers and community members from Zee Aing Village.

2.3 Environmental Impact

Since early 2016 the Environmental Impact Assessment (EIA) of 170 MW Capacity Minbu Solar Power Farm had been conducted and the study was approved in mid-2017 by Environmental

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Conservation Department (ECD) of Ministry of Natural Resource and Environmental Conservation Department, Myanmar. The report has been prepared in accordance with acting EIA procedures and requirement endorsed by the ministry. The project owner commissioned independent third party, Myanmar Survey Research Co., Ltd. to assess the environmental effects of the all the project activities/actions with respect to project different project stages (i.e. pre-construction, construction, operation and maintenance, and dismantling). EIA concludes that the project does no net harm to the environment. The key environmental impacts are summarized as follow:

- The project reduces CO2 emissions since it reduces the amount of fossil fuel used. In the case of “no project”, the stated amount of electricity would be generated from fossil fuels and cause air pollution.
- The project reduces fly ash emissions since it reduces the amount of fossil fuel used. In case of “coal- fired power plant”, stated amount of electricity would be generated from fossil fuels and cause air pollution.
- The solar power system does not need a treatment system. The washed water for the PV surface is totally harmless.

2.4 Public Comments

The main comments were concerns for land compensation and how the project would benefit the local community livelihoods during the construction and operation phases.

The land compensation case will be officially conducted for the local communities with the joint effort with the local stakeholders. Besides, the productive and effective CSR program will be implemented and developed for the local community and neighbouring communities.

2.5 AFOLU-Specific Safeguards

Not applicable

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

ACM0002: “Grid-connected electricity generation from renewable sources” (Version 21.0);

The methodology also refers to:

“Tool for the demonstration and assessment of additionality” (Version 07.0.0);

“Tool to calculate the emission factor for an electricity system” (Version 07.0);

“Investment analysis” (version 11.0)

More information concerning the above methodology and tools can be referred to:

http://cdm.unfccc.int/methodologies/PAmethodologies/approved.html

3.2 Applicability of Methodology

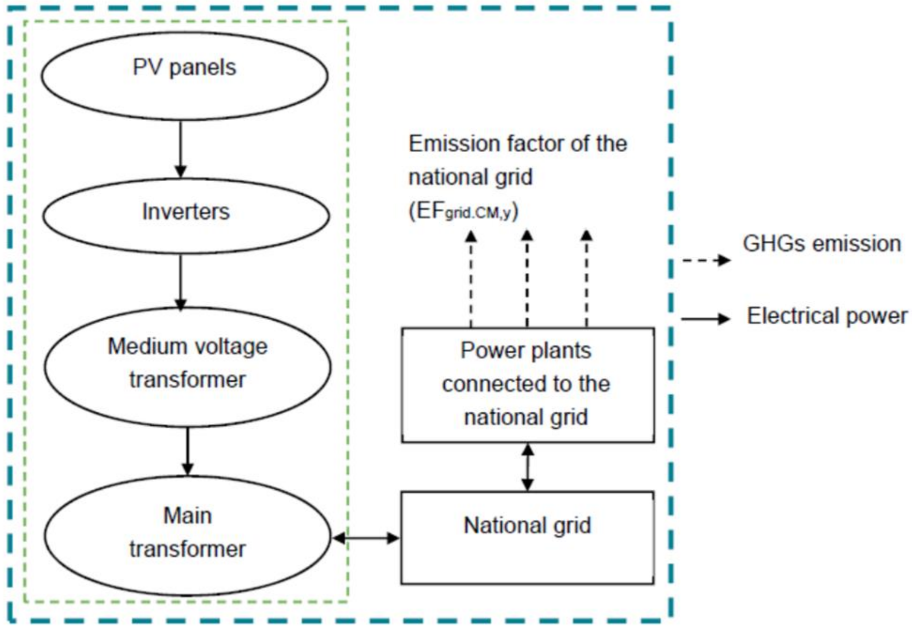
The approved methodology ACM0002 is applicable to the project activity, because:

- The Project is a newly built 220 MWp grid-connected renewable energy (i.e. photovoltaic power) power plant at a site where no renewable power plant was operated prior to the implementation of the project activity. Therefore, the baseline methodology ACM0002 (version 21.0) is applicable to the project activity.
- In addition, project activity will supply electricity to the EPGE, so “Tool to calculate the emission factor for an electricity system” can be used to estimate OM, BM and/or CM for baseline emission calculation.

3.3 Project Boundary

The project boundary includes the solar power plant, sub-stations, grid, and all power plants connected to the national grid. The proposed project activity will evacuate power to the Myanmar national grid. Therefore, the entire Myanmar national grid and all connected power plants have been considered in the project boundary for the proposed project activity.

The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions.



The project does not involve any other emissions sources not foreseen by the methodologies. The greenhouse gases and emission sources included in or excluded from the project boundary are shown in table below. The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions.

Source		Gas	Included?	Justification/Explanation
Baseline	Grid Connected Electricity Generation	CO2	Yes	Main Emission source
		CH4	No	Minor Emission source
		N2O	No	Minor Emission source
		Other	-	-
Project	Greenfield Solar Power Project activity	CO2	No	No CO2 emissions are emitted from the project
		CH4	No	Project activity does not emit CH4
		N2O	No	Project activity does not emit N2O
		Other	-	-

3.4 Baseline Scenario

The project is newly built power plant in Myanmar so the baseline scenario for greenfield power plant is applied. The baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07.

3.5 Additionality

The additionality of the proposed project was demonstrated in accordance with Tool for the demonstration and assessment of additionality version 7.0. The steps of consideration are as follow:

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

The project is constructed under the laws and regulations of Myanmar. The project activity undertaken without being registered as a CDM project activity.

Step 2: Investment analysis

Sub-step 2a: Determine appropriate analysis method

Tool for the Demonstration and Assessment of Additionality (version 07.0) recommends three analysis methods, namely simple cost analysis (Option I), investment comparison analysis (Option II) and the benchmark analysis (Option III). The project is the first solar power plant in Myanmar so alternative comparison cannot be applied. Option I and option II is not appropriate. The benchmark analysis method (Option III) is the most appropriate option. It will be analysed in the Sub-step 2b below.

Sub-step 2b: Apply benchmark analysis

According to the Methodology tool of Investment Analysis, version 11.0 section 6 paragraph 15, “Local commercial lending rates or weighted average costs of capital (WACC) are appropriate benchmarks for a project IRR”. According to the World Bank, the lending interest rate in Myanmar is 16%. So, the lending rate of 16% is used as a benchmark for project IRR.

Table 1 Main assumptions

Power Plant - Main Parameters	
Power Plant Net Capacity (MW)	50
Energy Production yield	1,600
Tariff (USD)	0.1275
Degradation Rate	0.67%
Project Operation Life Cycle (years)	30
MMK/USD Exchange Rate	2,101.07
Benchmark IRR	16.0%

The project supplies electricity to EPGE for a 30 years' period ending 26 February 2049 with an agreed tariff rate of USD 0.1275. Phase 1 operated on 27th September 2019. It has a capacity of 50MW(DC).

According to the Methodology tool of Investment Analysis, version 11.0 section 5 paragraph 13, "the cost of financing expenditures shall not be included in the calculation of project IRR", so debt financing is excluded in the calculation. The IRR is calculated using post-tax cash flow. The result of the calculation is shown in table 2 below.

Table 2 Financial indicator comparison

	Proposed Project	Benchmark
IRR	5.99%	16.00%

By using the main assumptions stated in Table 1, proposed project IRR (post tax) of the proposed project is 5.99% which is lower than the benchmark of 16% which means that the project is not financially attractive.

Sensitivity analysis

Variation %	- 10%	Normal	+ 10%	Variation to reach benchmark
Tariff	4.85%	5.99%	7.84%	44%
Investment	7.42%	5.99%	5.51%	-103%
Operating Cost	6.76%	5.99%	5.20%	N.A

Sensitivity has been carried out for the 3 main assumptions: revenue (tariff), investment cost and operating cost. Variations for the sensitivity analysis is + 10% from the base case scenario. Variation to reach the benchmark is also carried out to see how much each assumption needs to be in order to reach the benchmark.

Results from the sensitivity analysis shows that even with the +10% of the main assumptions the IRR of the proposed project does not reach the benchmark of 16%. In order for revenue (tariff) to reach the benchmark, it needs to increase by 44% to USD 0.2264. While investment cost has to be reduced by 103% from the current investment to USD 35,959,105.

Step 3: Barrier Analysis

Not applicable

Step 4: Common Practice Analysis

Not applicable. The project is the first solar power plant operated in Myanmar.

3.6 Methodology Deviations

There is no deviation applied to this project

4 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

According to consolidated Methodology ACM0002 version 21.0, baseline emissions include only CO2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants.

The baseline emissions are to be calculated as follows:

$BEy = EGPJ,y \times EFgrid,CM,y$ Equation (11)

Where:

BEy = Baseline emissions in year y (t CO2/yr)

$EGPJ,y$ = Quantity of net electricity generation that is produced and fed into the grid

as a result of the implementation of the CDM project activity in year y

(MWh/yr)

$EFgrid,CM,y$ = Combined margin CO2 emission factor for grid connected power

generation in year y calculated using TOOL07 (t CO2/MWh).

Calculation of combined margin CO2 emission factor

the data is available from the IFI Default Grid Factor provided that Myanmar has $EFgrid,CM,y$ of 0.6202 (https://unfccc.int/documents/461676)

4.2 Project Emissions

PEy = 0

4.3 Leakage

As per ACM0002, no leakage emissions are considered.

4.4 Estimated Net GHG Emission Reductions and Removals

Year	Estimated baseline emissions or removals (tCO2e)	Estimated project emissions or removals (tCO2e)	Estimated leakage emissions (tCO2e)	Estimated net GHG emission reductions or removals (tCO2e)
Year 1 ((27/09/2019-31/12/2019))	12,181	0	0	12,181
Year 2	48,722	0	0	48,722
Year 3	48,722	0	0	48,722
Year 4	48,722	0	0	48,722
Year 5	48,722	0	0	48,722
Year 6	48,722	0	0	48,722
Year 7	48,722	0	0	48,722
Year 8	48,722	0	0	48,722
Year 9	48,722	0	0	48,722
Year 10	48,722	0	0	48,722
Year 11 (01/01/2028-26/09/2028)	36,542			36,542
Total	**487,220**	**0**	**0**	**487,220**

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EFgrid,OM ,y
Data unit	t CO2/MWh
Description	Simple operating margin emission factor
Source of data	IFI Default Grid Factors 2021 v3.1 from UNFCCC[1]
Value applied:	0.602
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	To calculate emission factor of the grid
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	EGm,y, EGy, EGk,y and EGn,h
Data unit	MWh/Year
Description	Net electricity generated by (Solar)power plant/unit m, k or n (or in the project electricity system in case of EGy) in year y or hour h
Source of data	Electricity meter readings on-site
Description of measurement methods and procedures applied	Estimated annual generation forming the basis for emission reduction calculation is 80,934 MWh as indicated in generation license.
Frequency of monitoring/recording	Monthly
Value applied:	-

Monitoring equipment	Type of meter	ITRON-SL7000
	Location of meter	On-site
	Accuracy of meter	0.5S
	Serial number of meter	There are 4 meters: 1: 83745225 2: 83745226 3: 83745227 4: 83745228
	Calibration frequency	Yearly
	Date of Calibration/ validity	Dec 2022
	Reference No. of Calibration Certificate	-
	Calibration Status	Not yet
QA/QC procedures applied	-	
Purpose of data	To calculate the baseline emission value	
Calculation method	-	
Comments	-	

5.3 Monitoring Plan

Monitoring process is to guarantee that the collected data is accurate and measurable to emission reduction from the proposed project. Net electricity generation is measured and recorded for billing purposes. Therefore, no new additional protocol is needed for monitoring emission reduction. The Plant Manager is responsible for the electricity generated, gathering all relevant datas and keeping the records.

Team Members are including the following staffs :

- Operation manager: Responsible for collecting data and preparing data for other team members
- Deputy plant manager Responsible for checking the number
- Plant manager: Responsible for approving the number
- Mann sub-station plant manager: Responsible to cross check the number and approve the number

Installation of meter and data monitoring are carried out according to the regulations by ESIA. Four metering devices (Two main and two spare devices) used for monitoring the electricity generated by the power plant. Readings are done using main metering devices and spare metering devices are used for comparison only. Data from metering devices is recorded by the project owner monthly (through remote reading). The data collected is witnessed by Mann sub-station representatives.

Four calibrated meters back up each other. Maintenance and calibration of the metering devices are made by EPGE. If there is a significant difference between the readings of two devices, maintenance and tests of the metering devices and the associated equipment are done before waiting for the periodical maintenance. The meters should comply with the regulation which define the accuracy class of the meters as 0.2 or 0.5 depending on the capacity of the circuit. EPGE records will be taken in consideration while calculating net electricity generation by the plant.

All data is kept for at least two years after the crediting period for QA/QC purposes. Calibration of the metering devices is made by EPGE and sealed before the commissioning of the power plant. The meters are calibrated by EPGE when there is an inconsistency between two devices.

6 ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS

6.1 Data and Parameters Monitored

Data / Parameter	EGm,y, EGy, EGk,y and EGn,h
Data unit	MWh
Description	Net electricity generated by (Solar)power plant/unit m, k or n (or in the project electricity system in case of EGy) in year y or hour h
Value applied:	152,884 MWh
Comments	-

6.2 Baseline Emissions

According to consolidated Methodology ACM0002 version 21.0, baseline emissions include only CO2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants.

The baseline emissions are to be calculated as follows:

$BEy = EGPJ,y \times EFgrid,CM,y$ Equation (11)

Where:

BEy = Baseline emissions in year y (t CO2/yr)

$EGPJ,y$ = Quantity of net electricity generation that is produced and fed into the grid

as a result of the implementation of the CDM project activity in year y

(MWh/yr)

$EFgrid,CM,y$ = Combined margin CO2 emission factor for grid connected power

generation in year y calculated using TOOL07 (t CO2/MWh).

Calculation of combined margin CO2 emission factor the data is available from the IFI Default Grid Factor provided that Myanmar has $EFgrid,CM,y$ of 0.6202 (https://unfccc.int/documents/461676)

6.3 Project Emissions

PEy = 0

6.4 Leakage

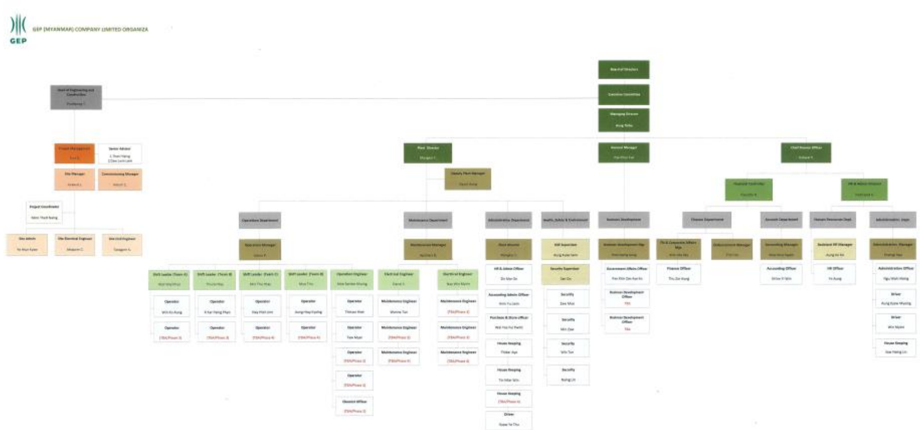
As per ACM0002, no leakage emissions are considered.

6.5 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
2019 (Sept – Dec)	12,890	-	-	12,890
2020	48,722	-	-	48,722
2021	46,632	-	-	46,632
2022 (Jan - Nov)	44,641	-	-	44,641
Total	152,884			152,884

APPENDIX 1: <EVIDENCE OF SUSTAINABLE DEVELOPMENT CONTRIBUTIONS >

Organization strcuture



CSR plan for the Lay Pin Village High school

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CSR Proposal
Lay Pin Village High School Electricity 5KV Solar roof top

Executive Summary

Lay Pin High school is located in Lay Pin village (6 kilometers far from MSPP), there are about 280 students and 9 teachers in this school. There is no electricity supply from Government.

The total area of school is 9.63 acres. Their compound has total 6 buildings. 4 buildings are class room, 1 building is class room and office room and one building is teacher housing room. One building install lighting cable but not yet use.
There have total 13 room, 9 class rooms for student, 3 rooms for teacher housing and 1 room for school office. There have total 6 toilets; 4 toilets for student and 2 toilets for teacher.

There have only 1 small generator supply for Printer and Computer of a teacher when they need to use only for important document have to submit Government headquarter or another importance needed.
They have purified drinking water system but can't use because there is no electric supply

Tribulations.

1. They can't use purify drinking water system because there has no electric power supply. Now, they are drinking only normal water. That's not purify water. It's not good for health for long time.
2. Before exam periods, school open at night time for studying lecture (Especially for high school) because that village no have electricity in these conditions. So, they are facing a little difficult without electricity long time.
3. In the current condition, all the class room are poor ventilation because no ventilation fan.
4. Most of teacher come from other region and stay at the school housing. So, they need the electricity for lighting, water supply pump and cooking for more comfortable.
(As per teacher information)

Advantage

From the village and school side of view,
1. The student will learn comfortable more than before.
2. They can drink purify drinking water for their healthy
3. The teacher can teach and stay comfortable more than before.

From the Company side of view,
1. We will more good reputation / image for our Company
2. We will be a part of community development around the power plant and have a good relationship with each other in a sustainable way.
3. MSPP is a part of Lay Pin Village due to the positive effect, they will be thankful from their hearts and it will be good cooperation and better relationship with our Company.